GrowthStreak: Product Requirements Document and Implementation Plan

Flutter app for daily goals, friend-verified fitness, step counter and penalty wallet. Final version.

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PART A. PRODUCT REQUIREMENTS DOCUMENT

A1. Overview

Problem. Students set daily goals but have no visual record of growth and no real accountability, especially for fitness.

Solution. A goal tracker with percentage-based daily, monthly and yearly progress, category growth graphs, an automatic step counter, and a friend-verified gym check-in. If a verified task is missed and the friend does not accept a reason, a penalty (default Rs 10) is paid to the friend from a pre-loaded wallet.

Target user. Engineering students aged 17 to 25.

Success metrics. At least 70% daily task completion. At least 90% of proofs decided within 2 hours. Zero duplicate or failed penalty transfers. 30-day retention above 40%.

A2. Categories

Category	Examples
Health	Gym, steps, water, sleep
Academics	Study hours, assignments, exams
Technical Skills	DSA, Flutter, certifications
Projects	Milestones and monthly updates
Custom	User-defined

A3. Feature scope (final)

#	Feature	Verification	Penalty
1	Daily tasks with weighted percentage, calendar, month and year totals	Self-marked	Optional
2	Categories and	Self-marked	Optional

#	Feature	Verification	Penalty
	custom categories		
3	Gym task with photo	Automatic WhatsApp approval by the friend	Rs 10 default
4	Missed-day reason	Friend accepts or declines on WhatsApp	Cancelled only if accepted
5	Step counter	Automatic from Health Connect, HealthKit or phone sensor	Optional
6	Projects with monthly updates	Self-written	None
7	Analytics: bar, line, donut charts, streaks, excused days	Computed	None
8	Wallet: top-up, hold, charge, weekly payout to friend	Payment provider	n/a
9	Notifications and reminders	n/a	n/a
10	Add, edit, pause, delete tasks	n/a	n/a

A4. One accountability friend

Each user has exactly one accountability friend who handles all tasks needing approval or a penalty. The friend must opt in once by messaging the business WhatsApp number with a one-time code. The friend's number must differ from the user's. The friend can be changed once every 30 days; proofs already sent stay with the old friend until decided. If the friend sends STOP, Gym and penalty tasks pause and no penalty is charged until a new friend opts in. If messages fail for 3 days, penalties are voided and the user is asked to choose a new friend.

A5. Core user flows

Gym photo approval (fully automatic)

1. User opens the Gym task and takes a photo in the app (camera only).
2. App compresses and uploads it to private storage.
3. Backend sends the friend a WhatsApp template message with the photo and two buttons: Approve and Reject.
4. Friend taps a button. Meta calls the webhook.
5. Backend verifies the signature, the sender number and that the proof is still pending, then updates the proof, completes the task and recomputes stats.

6. App updates the Health bar within seconds through a realtime listener.

Rejected: the task stays pending and the user may retake the photo (maximum 3 attempts per task per day).

Missed-day reason

1. The gym task is open and the user cannot attend. They tap Send a reason (10 to 300 characters, optional category).
2. The penalty is placed on hold. The friend receives the reason with Accept and Decline buttons.
3. Accept: penalty cancelled, task marked Excused. Decline: penalty charged. The friend can also type YES or NO within the 24-hour window.

Limits: one reason per task per day, maximum 3 accepted excuses per month, submitted before the grace deadline (default 10:00 AM next day).

Decision table

Situation	Penalty	Task status
Gym photo approved	None	Completed
Reason accepted	None	Excused
Reason declined	Rs 10 charged	Missed
Reason sent, friend never reads it	None	Voided
Reason sent, friend reads it, no answer by grace deadline	Rs 10 charged	Missed
No photo and no reason by grace deadline	Rs 10 charged	Missed
Message could not be delivered	None, fallback offered	Voided
Steps goal reached	None	Completed
Steps goal not reached at grace deadline (penalty enabled)	Rs 10 charged	Missed

Excused days neither add to nor break the streak, and excused tasks are removed from the day's total weight.

Step counter

Daily goal default 8,000 (range 1,000 to 50,000). Steps come from Health Connect (Android) or HealthKit (iOS) through the health package, with the phone step sensor as a fallback. The app syncs on open, on resume and every 15 minutes in the background. The task completes automatically when steps reach the goal. Manually entered records are ignored; days above 60,000 steps or sustained rates above 200 steps per minute are flagged for review. Distance uses height times 0.415 as stride; calories are a labelled estimate.

Wallet and penalty

The user tops up the wallet by UPI. A penalty first goes on hold, then is charged after the grace deadline with an idempotency key (taskId and date) so it can never be charged twice. Charged penalties are paid to the friend in a weekly batch to reduce fees.

A6. Screens and navigation

Bottom navigation: Home, Analytics, Tasks, Wallet, Profile.

Screen	Key content
Splash and onboarding	Illustration, tagline, Get Started
Home	Greeting, green hero card with today's percentage and change, month and year chips, quick actions, category progress, today's tasks
Calendar	Month grid with colour rings (green done, amber partial, red missed, blue excused), month and year totals
Analytics	Week, Month, Year toggle, bar chart, line chart, donut by category, top categories, streaks
Tasks	Monthly completion card, filters, Add Task
Add or edit task	Title, category, type, repeat, reminder, weight, penalty
Gym proof	Camera, preview, status timeline: Sent, Delivered, Read, Decision
Missed-day reason	Category chips, text box, send
Step detail	Ring, hourly chart, distance, calories, goal streak
Projects	Active and Completed tabs, project cards, monthly update timeline
Wallet	Balance card, Add Money, held penalties, transaction history
Profile	Discipline score ring, friend and opt-in status, notifications, settings, delete account

Visual style. Deep emerald brand colour (#0B5D45, light #E6F4EE), background #F5F9F7, white cards with 16 to 20 px radius and soft shadows, Poppins or Inter, tinted icon tiles per category, success #22A06B, warning #F5A524, danger #E5484D. Reusable components: HeroCard, StatTile, CategoryTile, TaskTile, StatusChip, SegmentedTabs, ProgressBar, DonutChart, BarChartCard, LineChartCard, PrimaryButton, BottomNav.

A7. Required input fields

Profile

Field	Required	Validation
Name	Yes	2 to 50 characters
Email or Google account	Yes	Valid email
Mobile number	Optional	E.164, OTP verified
Height and weight	Optional	100 to 250 cm, 30 to 250 kg
Daily step goal	Optional	1,000 to 50,000
Profile photo	Optional	Image up to 2 MB

Task

Field	Required	Validation
Title	Yes	2 to 80 characters
Category	Yes	From the list
Type	Yes	Standard, Gym (photo), Steps (auto)
Repeat	Yes	Once, daily, selected weekdays
Reminder time	Optional	Valid time
Weight	Optional	1 to 10, default 1
Penalty amount	Optional	Rs 0 to Rs 500, default Rs 10
Start and end date	Optional	End after start

Friend

Field	Required	Validation
Name	Yes	2 to 50 characters
WhatsApp number	Yes	E.164, not the user's own
UPI ID	For payouts	Valid VPA

Gym proof: photo (camera only, up to 2 MB, taken within 30 minutes), optional caption up to 100 characters.

Missed-day reason: reason text 10 to 300 characters, no links, optional category, within the grace window and under the monthly cap.

Project update: title 2 to 80, description up to 500, deadline in the future, update text 1 to 1,000, progress 0 to 100, up to 3 images of 2 MB each.

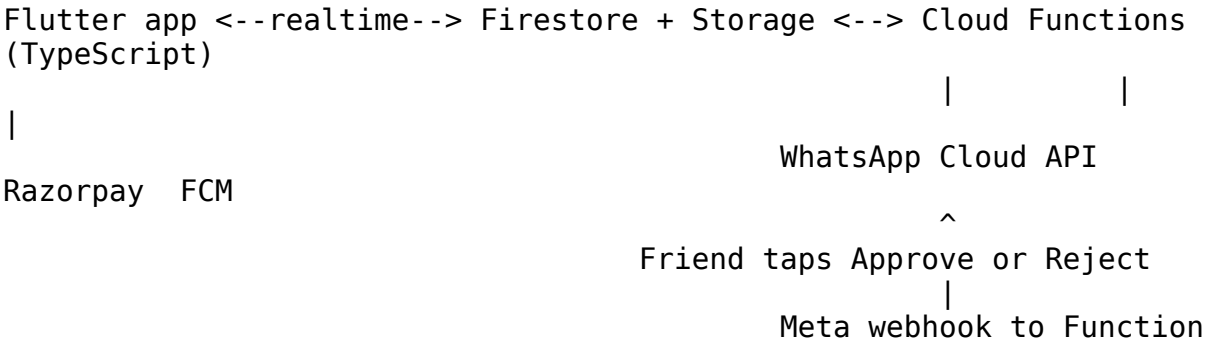
Wallet: top-up Rs 50 to Rs 5,000 per payment via UPI; withdrawal up to available balance.

A8. Non-functional requirements

Area	Target
Gym photo to friend's WhatsApp	Under 10 seconds
Friend decision to app update	Under 5 seconds
Step sync freshness	At most 15 minutes
App cold start	Under 3 seconds
Charts	60 fps
Offline	Tasks tick offline and sync later
Availability	99.5% for functions and webhooks
Platform	Android first, iOS later

PART B. IMPLEMENTATION PLAN

B1. Architecture



B2. Frontend (Flutter)

Structure. Feature-first folders: lib/core (theme, widgets, utils), lib/features/{home, calendar, analytics, tasks, steps, proofs, projects, wallet, profile}, lib/data (models, repositories). Riverpod for state, go_router for navigation, Freezed and json_serializable for models, repositories behind interfaces for testing. Two flavors, dev and prod.

Packages. firebase_core, firebase_auth, cloud_firestore, firebase_storage, cloud_functions, firebase_messaging, firebase_app_check, camera, flutter_image_compress, health, pedometer, workmanager, fl_chart, table_calendar, razorpay_flutter, flutter_local_notifications, hive, connectivity_plus, google_fonts, flutter_animate.

Realtime behaviour. Home listens to today's day log and tasks. Gym proof listens to its proof document and shows the status timeline. Wallet listens to the balance. Calendar and Analytics read pre-computed month and year statistics.

Offline and quality. Tasks are written locally first and synced when online. Every screen has loading, empty and error states. Uploads retry with backoff. Images are compressed to about 300 KB. Dark mode, accessibility labels and pagination for history lists are required.

Permissions. Camera, notifications, Activity Recognition, Health Connect steps read. A prompt asks the user to exclude the app from aggressive battery saving so step sync works in the background.

B3. Backend (Firebase and Cloud Functions)

Module	Service
Authentication	Firebase Auth (Google and phone)
Database	Cloud Firestore
File storage	Firebase Storage (private bucket)
Server logic	Cloud Functions 2nd gen, TypeScript (Blaze plan with budget alert)
Scheduling	Cloud Scheduler
Push	Firebase Cloud Messaging
Messaging	WhatsApp Business Cloud API (Telegram bot as free fallback)
Payments	Razorpay (Route or Payouts)
Secrets	Secret Manager
Monitoring	Crashlytics, Cloud Logging, Cloud Monitoring

Functions

Function	Trigger	Job
sendProofToWhatsApp	Storage or Firestore create	Upload media, send template with buttons
sendExcuseToWhatsApp	Firestore create	Validate limits, hold penalty, send reason template
whatsappWebhook	HTTPS	Verify signature, handle opt-in, buttons, YES/NO text, STOP, delivery and read receipts
proofReminders, excuseReminders	Scheduler	Remind the friend after 2 and 6 hours
finalizeDay	Scheduler (every 15 minutes)	At grace deadline apply the decision table, charge or cancel held penalties
recomputeStats	Firestore trigger	Update day, month and year statistics and task outcomes
syncStepsTrigger	Firestore trigger	Complete the step task when

Function	Trigger	Job
		goal is reached
createTopupOrder	Callable	Create Razorpay order
razorpayWebhook	HTTPS	Credit top-ups only after verified payment events, update payouts
weeklyFriendPayouts	Scheduler	Batch penalties and pay the friend
reconcilePayments	Scheduler	Compare provider settlements with the ledger
cleanup	Scheduler	Delete proofs older than 30 days
deleteAccount	Callable	Wipe user data

WhatsApp setup. Create a Meta Business account and developer app with the WhatsApp product. Use a phone number not already registered on WhatsApp. Complete business verification for real use. Submit utility templates for approval: gym_proof_request (image header, two quick-reply buttons), absence_reason_request (two buttons), proof_reminder and proof_result. Create a system user with a permanent token stored in Secret Manager. Configure the webhook and subscribe to message events. Free option for development: Meta's test number can message up to 5 pre-verified recipients. Unofficial automation of a normal WhatsApp account is not used because it risks a ban.

Webhook rules. Respond with 200 quickly, process idempotently using the Meta message ID, return 403 on an invalid signature.

B4. Database (Firestore)

Collection	Fields
users/{uid}	name, email, phone, timezone, heightCm, weightKg, stepGoal, fcmTokens, walletBalance (server only), accountabilityFriend {name, phoneE164 encrypted, upiId encrypted, optedIn, optedInAt, revokedAt, blockExcuses}, createdAt, deletedAt
users/{uid}/tasks/{id}	title, category, type, repeat, reminderTime, weight, penaltyAmount, startDate, endDate, active
users/{uid}/dayLogs/{date}	percent, completedWeight, totalWeight, byCategory, taskOutcomes, finalized
users/{uid}/stepLogs/{date}	steps, goal, hourly, distanceM, kcal, source, manualEntryDetected, flagged, lastSyncAt

Collection	Fields
users/{uid}/monthStats/{yyyy-mm}, yearStats/{yyyy}	avgPercent, daysTracked, excusedDays, excusesUsed, steps, byCategory
proofs/{id}	uid, taskId, friendPhone, photoPath, photoHash, status, waMessageId, attempt, sentAt, deliveredAt, readAt, decidedAt, decision, expiresAt
excuses/{id}	uid, taskId, friendPhone, date, reason, category, status, waMessageId, sentAt, readAt, decidedAt, graceUntil
walletTxns/{id}	uid, type (topup, hold, charge, cancel, payout, refund), amount, status, idempotencyKey, razorpayRef, relatedTaskId, createdAt
users/{uid}/projects/{id}/updates/{yyyy-mm}	title, description, deadline, status, progress, text, attachments
waEvents/{metaMessageId}	processedAt (deduplication)
auditLog/{id}	actor, action, ref, timestamp

Status machines. Proof: uploading, sent, delivered, read, approved or rejected or expired or failed. Excuse: submitted, sent, delivered, read, approved or declined or expired or failed. Task outcome: pending, completed, excused, missed, voided.

Indexes. proofs by (uid, status, expiresAt); walletTxns by (uid, createdAt descending); tasks by (uid, active).

Retention. Gym photos deleted 30 days after the decision. Ledger records kept 7 years.

B5. Security

Firestore rules (summary). Users read and write only their own documents and cannot modify walletBalance. Clients may create a proof or excuse only with the initial status and valid fields; all later changes, decisions, penalty state and the whole wallet ledger are written only by Cloud Functions. Everything else is denied.

Storage rules. Private bucket, image types only, 2 MB maximum, user can write only to their own path.

Area	Control
Authentication	Firebase Auth, verified email or phone, session revocation
App integrity	Firebase App Check (Play Integrity) enforced on Firestore, Storage and Functions
Approval integrity	Decision accepted only from the opted-in friend's number, with a valid Meta signature, a pending and unexpired proof

Area	Control
Photo anti-cheat	Camera only, server timestamp within 30 minutes, perceptual hash to reject reused photos, maximum 3 attempts per day
Step anti-cheat	Ignore manual entries, flag extreme values, record the data source
Collusion limits	Friend cannot be the user's own number; weekly approve and decline summary to both
Money	Balance changes only in Functions, atomic transactions, idempotency key taskId_date, append-only ledger, daily reconciliation
Secrets	Secret Manager for WhatsApp, Razorpay and webhook secrets; never in the app or in Git
Transport and storage	HTTPS only; signed photo URLs that expire in minutes; encrypted phone and UPI fields
Abuse	Rate limits on callable functions, per-user WhatsApp message caps, excuse limits
Privacy	Consent screen for photo sharing and health data, STOP honoured immediately, privacy policy, data deletion, India DPDP Act compliance
Store compliance	Health Connect permissions declaration and Data Safety form
Monitoring	Alerts for webhook errors, signature failures, payout failures and spend spikes; audit log of state changes

B6. Payments (production)

1. Register a business entity (a sole proprietorship with PAN and a current bank account is usually enough) and complete live KYC with Razorpay or Cashfree. Publish Terms, Privacy, Refund policies and a contact page.
2. Choose a money-movement model with the provider before building: Razorpay Route (penalties transferred to the friend's linked account after the friend's KYC), RazorpayX Payouts from a prefunded balance to the friend's UPI, or a UPI AutoPay mandate with no wallet.
3. Confirm in writing with the provider or a CA whether holding user funds falls under RBI payment aggregator or prepaid instrument rules. For you and a few friends, start small.
4. Top-up flow: createTopupOrder, user pays through checkout, credit only after the verified payment.captured webhook.
5. Ledger: every top-up, hold, charge, cancel, payout and refund is a row; the balance is derived and checked daily.

6. Fees on Rs 10 are significant, so batch friend payouts weekly. Failed payouts retry with backoff and alert you.
7. Provide withdrawal of unused balance, a dispute form and manual review log. Check GST and receipts with a CA.
8. Test with test keys, then with real small amounts (Rs 1 to Rs 10) on your own accounts before going live.

B7. Cost

Item	Free option	Paid
Firebase Auth (email, Google)	Free	Phone OTP SMS costs a few rupees each
Firestore, Storage, Hosting	Free monthly allowance	Small usage fees
Cloud Functions	Needs Blaze plan, free allowance included	Usually near zero at small scale; set a budget alert
FCM	Free	None
WhatsApp Cloud API	Meta test number (5 recipients)	Per-message template charges once live; verify India rates
Telegram bot alternative	Free	None
Razorpay	Test mode	Per-transaction and payout fees
Domain	Free web.app address	About Rs 800 to Rs 1,000 per year
Google Play	None	One-time fee of about 25 USD
Apple Developer	Not needed for Android first	About 99 USD per year

Prices change often; check each provider's current pricing page.

B8. Environments, CI/CD and deployment

Environments. growthstreak-dev, growthstreak-staging and growthstreak-prod Firebase projects, with separate Meta test app and Razorpay test keys outside production. Firebase Emulator Suite for local work. GitHub Actions for lint, tests, build and deploy. Daily Firestore export for backups.

Backend deploy. Create the prod project, enable the Blaze plan with a budget alert, add all secrets, then run: firebase deploy –only firestore:rules,firestore:indexes,storage,functions,hosting. Point the WhatsApp and Razorpay webhooks at the production function URLs.

App release. Build with flutter build appbundle –release –flavor prod using a signed upload keystore (back it up). Create a Google Play developer account, complete the store listing, privacy policy URL, Data Safety form and content rating. Start with internal testing. New personal developer accounts usually need a closed test with about 12 testers for 14 days before

production access; confirm the current Play Console rules. Then roll out in stages of 10%, 50% and 100% while watching Crashlytics.

Go-live gate. Razorpay live mode approved. WhatsApp templates approved and number verified. Staging passes photo approval, reason accept, reason decline, no reply and failed delivery. A real Rs 10 top-up, hold, penalty and payout completes and reconciles. Security-rule tests pass and no secrets are in the repository. Alerts, backups and an incident contact are set.

B9. Test plan and acceptance criteria

Test	Pass condition
Gym flow end to end	Photo reaches the friend within 10 seconds; Approve updates the app within 5 seconds
Rejection	Task stays pending; resend works; limit of 3 enforced
Reason accepted	Penalty cancelled, task Excused, percentage excludes the task
Reason declined	Penalty charged once after the grace deadline
Excuse limit	The 4th excuse in a month is blocked
Duplicate or late taps	Only the first valid decision is applied
Wrong sender	Reply from an unregistered number is ignored and logged
Step sync	Steps within 15 minutes; manual entries ignored; goal completes the task
Offline	Tasks tick offline and sync correctly
Penalty	Exactly one Rs 10 charge per missed task even if the scheduler runs twice
Real payment	Top-up credited only after the verified webhook, with no double credit on retry
Failed payout	Retries, then alerts; ledger stays consistent
Security	Client cannot change walletBalance, proofs, excuses or other users' data
Load	1,000 simulated users at day end with no missed or double penalties
Recovery	Webhook outage replays correctly after restore
Reconciliation	Zero mismatch for 7 consecutive days

Definition of done. All tests pass in staging with real WhatsApp test numbers, a 7-day beta with no money errors, and verified monitoring alerts.

B10. Delivery timeline

Phase	Scope	Time
1	Setup, theme, Auth, rules, base components	1 week
2	Tasks, percentages, calendar, analytics	2 weeks
3	Step counter with Health Connect and background sync	1 week
4	WhatsApp setup, opt-in, gym proof, webhook	2 weeks
5	Missed-day reason, grace logic, finalizeDay	1 week
6	Wallet, Razorpay test mode, ledger, penalties	2 weeks
7	Projects and polish	1 week
8	Security review, load tests, closed beta	2 weeks
9	Live payments, store release, staged rollout	2 to 3 weeks

Realistic total: about 14 to 16 weeks for one developer, including waits for Meta, Razorpay and Google Play approvals.

B11. Risks

- Payment and wallet regulation: use a licensed partner and confirm the structure in writing.
- Friend does not respond: reminders, a grace window, and no penalty if the message was never read.
- Fake photos: camera-only capture, timestamp and hash checks, and friend verification.
- Fees large relative to Rs 10: weekly batching.
- Approval delays from Meta, Razorpay and Google: start these processes in week 1.